SADC HUMANITARIAN AND EMERGENCY OPERATIONS CENTRE

TRAINING MANUAL

ICT, Data Management and Reporting

Data Management | ICT Infrastructure | Reporting | BCM & DR | Cybersecurity | Steady-State ICT

Based on SOP Volume 6, Version 6.0 (Consolidated — Full ICT Policy Compliance)

Classification:	CONFIDENTIAL
Version:	1.0
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SOP Reference:	Volume 6, v6.0 (SOPs 6.1–6.6)
Annexes:	58 Operational Templates (6.1-A to 6.6-I)
Target Audience:	ICT Officers, IM Officers, Data Analysts, System Administrators

Benchmarked: ISO 27001:2022 | ISO 22301:2019 | UNDAC 8th Ed. | IFRC Surge Toolbox | WFP ETC | SADC NETP/PPDR

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1. Introduction and Purpose

This Training Manual is a companion to SHOC SOP Volume 6: ICT, Data Management and Reporting, Version 6.0. It provides ICT Officers, IM Officers, Data Analysts, and System Administrators with practical, SOP-by-SOP guidance on the six interdependent ICT procedures structured per the UNDAC Mission Cycle.

The six SOPs form an integrated ICT ecosystem: **SOP 6.1 (Data Management), SOP 6.2 (ICT Infrastructure), SOP 6.3 (Assessment & Reporting), SOP 6.4 (BCM & Disaster Recovery), SOP 6.5 (Cybersecurity Operations), and SOP 6.6 (Steady-State ICT Operations).**

Key principle: SHOC's ICT resilience extends beyond crisis response to permanent organisational capability. SOPs 6.4–6.6 ensure SHOC survives its own ICT disruptions, not just responds to external disasters.

2. Terms and Abbreviations

Abbreviation	Full Form
BCM	Business Continuity Management
CAP	Common Alerting Protocol
DLP	Data Loss Prevention
DRaaS	Disaster Recovery as a Service
DRMIMS	Disaster Risk Management Information Management System
EDR	Endpoint Detection and Response
ETK	Emergency Telecommunications Kit
IR	Incident Response (cybersecurity)
MDM	Mobile Device Management
PPDR	Public Protection and Disaster Relief
RBAC	Role-Based Access Control
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SIEM	Security Information and Event Management
SITREP	Situation Report
SOC	Security Operations Centre
SPOF	Single Point of Failure
TESS+	UN Telecommunications Security Standards

3. How to Use This Manual

Each SOP section follows the UNDAC Mission Cycle (Preparedness, Alert, Activation & Mobilisation, Operations, Demobilisation & Transition). SOPs 6.4–6.6 operate continuously across all phases.

3.1 Operational Cycle by SOP

Phase	SOP 6.1 Data	SOP 6.2 Infrastructure	SOP 6.3 Reporting
Preparedness	Verify feeds; test validation; metadata baselines	Monthly SATCOM tests; maintain VPNs; patch management	Quarterly risk assessments; baseline dashboards
Alert	Heightened QA; verify surge data quality	Verify MNO capacity; pre-check satellite; heightened SOC	Daily SITREPs; 6-hour dashboards
Activation	Full real-time ingestion; partner feeds active	Deploy ETK; activate Starlink/VSAT; full SATCOM	3–6 hourly SITREPs; Flash Updates; real-time dashboards
Operations	Real-time validation; 24/7 data flow; continuous backup	24/7 network ops; enhanced cybersecurity	Real-time dashboards; daily financial SITREPs; donor reports
Demobilisation	Archive per retention; secure disposal; lessons learned	Decommission field systems; equipment return; AAR	AAR within 30 days; update risk register; archive reports

3.2 Data Classification Scheme

Classification	Handling	Encryption	Retention
Public	No restrictions	Optional	Permanent
Restricted	SHOC staff only	In transit	5 years
Confidential	Need-to-know; encrypted	At rest + transit	7 years
Secret	Restricted access; audit trail	At rest + transit	10 years
Top Secret	ED approval; enhanced controls	Enhanced	Permanent

4. SOP 6.1: Data Management

SOP Reference: Section 5 (5.1–5.5)

SOP 6.1 governs data collection, validation, classification, storage, access control, and archiving for all SHOC information assets through DRMIMS.

Phase	Key Tasks	Annex
Preparedness	Verify all data feeds (DRMIMS, APIs, satellite, AMHEWAS, GDACS); test validation scripts; review classification policies; verify backup RPO compliance	6.1-A to 6.1-G
Activation/Ops	Full ingestion from all SOPs (≤1hr); apply metadata tags (≤2hr); automated validation + GIS cross-check (≤6hr); CAP message validation (≤30min); classify per 5-tier scheme; execute cloud backup per RPO	6.1-A to 6.1-F
Demobilisation	Archive per retention (5–10yr by classification); legal holds; secure disposal with certified destruction; permanent disposal log; data management lessons learned	6.1-J

4.1 Storage Architecture

Primary (Nacala servers) → Cloud Primary (Azure SA Region A, real-time replication) → Cloud Secondary (SADC Regional, geo-redundant DR) → Satellite Buffer (Botswana OSOCC, failover cache) → Cold Archive (off-site, 7+ year compliance).

4.2 Trainer's Notes

Critical exercise: Present a CAP message with deliberate errors (invalid hazard code, polygon outside SADC bounds, missing urgency field). Require trainees to execute validation per Step 2.3A within 30 minutes, flag non-compliant fields, and document in the Data Quality Form (Annex 6.1-F). Also test RPO compliance: simulate a 2-hour gap in EW data backup and require identification of the breach against the ≤15-minute EW RPO target.

5. SOP 6.2: ICT Infrastructure and New Technologies

SOP Reference: Section 4 (4.1–4.5)

SOP 6.2 deploys, manages, and optimises SHOC's technology ecosystem: cloud, satellite, terrestrial networks, cybersecurity baseline, and ETK. SHOC operates under PPDR Model III (commercial MNO + satellite backup).

Phase	Key Tasks	Timeline	Annex
Preparedness	Assess cross-functional ICT needs; monitor MNO performance monthly; monthly SATCOM tests; maintain SADC/AU VPNs; cybersecurity baseline (quarterly); ETK inventory verification	Annual/Monthly/Quarterly	6.2-A to 6.2-L
Alert	Verify MNO capacity; pre-check satellite; confirm AMHEWAS/GDACS feeds; heightened SOC; pre-position ETK; verify ERP and HRIS access	Upon EW threshold	6.2-A, 6.2-B
Activation	Activate backup circuits, scale bandwidth (≤2hr); deploy Starlink/VSAT (≤4hr); deploy ETK with ERT (≤4hr); activate DRaaS; coordinate WFP ETC if needed	≤4 hours	6.2-B, 6.2-I, 6.2-H
Operations	24/7 network ops; enhanced cybersecurity with SOC active; continuous satellite/terrestrial monitoring; real-time dashboards; RTO targets enforced	Continuous	6.2-K
Demobilisation	Decommission field terminals; recover ETK; infrastructure AAR; update lifecycle plans and vendor SLAs	Post-mission	6.2-I

5.1 PPDR Frequency Bands

SADC harmonised bands: Narrowband (380–430 MHz) for voice/telemetry; Wideband (380–470 MHz) for data/video; Broadband (698–791 MHz) for high-throughput field connectivity. ETK equipment shall conform to TETRA/DMR/P25 standards for cross-border interoperability.

6. SOP 6.3: Assessment, Data Analysis and Reporting

SOP Reference: Section 6 (6.1–6.6)

6.1 Reporting Framework

Report Type	Phase	Frequency	Audience
SITREP	Activation–Ops	3hr (crisis); 6hr (activation)	All stakeholders
Flash Update	Alert–Ops	Event-triggered	Leadership, NDMAs
EW Bulletin	All phases	Daily (prep); hourly (crisis)	NDMAs, Partners
Financial SITREP	Activation–Ops	Daily (per SOP 1.2)	ED, Donors
Donor Report	Ops–Demob	Weekly; Final	Donors, Secretariat
Dashboard Update	All phases	Real-time (crisis); daily (prep)	All users

6.2 SITREP Production Workflow

Aggregate data from all SOP domains (≤2hr) → Validate per SOP 6.1 (≤6hr) → Run analytical models and gap analysis (≤8hr) → Create visualisations and update dashboards (≤2hr) → Draft SITREP (10 sections, ≤3hr) → Quality review → HoP approval → ED sign-off (≤1hr) → Publish to SHOC Portal, DRMIMS, VOSOCC (≤30min).

6.3 Accessibility (NETP AI 10)

All public-facing products must be accessible: plain language (max 8th-grade level); high-contrast colour schemes for colour-blind users; sign language/captioning for audio; multilingual (EN/FR/PT + local languages); accessible tagged PDF format. Compliance verified per Annex 6.3-I before each publication.

7. SOP 6.4: Business Continuity and Disaster Recovery

SOP Reference: Section 7 (7.1–7.4)

SOP 6.4 ensures SHOC survives its own ICT disruptions through three-tier failover, DR testing, and power continuity.

7.1 Three-Tier Failover

Tier	Location	Activation	Threshold
Tier 1	Nacala on-site	Redundant path: secondary ISP (<30s), Starlink (<5min), generator/UPS for power	<30sec–5min
Tier 2	Botswana OSOCC	If Tier 1 fails after 30min or site evacuated: satellite uplink, redirect EW feeds, sync DRMIMS	<2 hours
Tier 3	Virtual OSOCC (Azure)	If Botswana also unavailable: cloud environment, VPN remote access, cloud-based EW/IM/Ops	<4 hours

7.2 DR Testing Schedule

Test	Frequency	Key Actions
Backup integrity verification	Monthly	Automated checksum; restore random system; RPO compliance check
Cloud restore drill	Quarterly	Restore DRMIMS + EW feeds to Azure secondary; verify data consistency; measure actual RTO
Satellite failover drill	Quarterly	Simulate ISP loss; verify auto-failover to Starlink within 30sec; test BGAN backup
Botswana OSOCC readiness	Quarterly	Verify uplink, DRMIMS sync lag <15min, data buffer, equipment, credentials
SPOF testing	Semi-annual	Disable each SPOF mitigation; verify failover; document remediation
Full BCM exercise	Annual	Simulate Critical site-loss; activate Tier 2/3; operate 4hr minimum; joint AAR with BCM team

7.3 Power Continuity

Monthly: UPS battery load test (30min sustained); verify automatic transfer switch (<10sec). Quarterly: generator full-load test (2hr); verify fuel ≥72hr at full load; test solar output. All tests documented in Power Continuity Test Log (Annex 6.4-G).

8. SOP 6.5: Cybersecurity Operations

SOP Reference: Section 8 (8.1–8.4)

8.1 Incident Response (5-Step)

Step	Action	Threshold
IR1: Detection	SIEM/SOC alert; classify: P1 (Critical), P2 (High), P3 (Medium), P4 (Low)	<15 min
IR2: Containment	P1/P2: isolate systems, disable accounts, preserve forensics. P3/P4: quarantine endpoint	P1: <1hr; P2: <4hr
IR3: Eradication	Remove malware; patch vulnerability; harden config; rotate credentials; external forensics for P1	<24hr (P1/P2)
IR4: Recovery	Restore from clean backup; validate data integrity per SOP 6.4; enhanced monitoring 72hr	Per RTO targets
IR5: Lessons Learned	AAR within 7 days; update SIEM rules; update training; semi-annual IR drill	7 days post

8.2 Vulnerability & Patch Management

Weekly vulnerability scanning (Critical remediation <24hr, High <72hr, Medium <14d, Low <30d). Annual penetration testing. Monthly patch cycle with 48hr staging test. Emergency patches (zero-day) deployed within 24hr with rollback plan.

8.3 Security Training

Quarterly awareness training: phishing simulation (target <10% click rate), password hygiene, social engineering, data classification. Annual ISO 27001 gap assessment by external auditor. New staff: mandatory completion within first week.

9. SOP 6.6: Steady-State ICT Operations

SOP Reference: Section 9 (9.1–9.6)

Function	Key Procedures	Annex
Daily Operations	0800 checklist: all systems operational, backup status, connectivity, SIEM alerts, UPS/generator; escalate RED items immediately	6.6-A
Help Desk	Ticket priorities: P1 (<1hr), P2 (<4hr), P3 (<8hr), P4 (<24hr); monthly SLA compliance report	6.6-B
Change Management	Standard (pre-approved); Normal (form → CAB → staging → deploy → verify); Emergency (immediate, document within 48hr)	6.6-C
Network/Cloud Monitoring	Latency thresholds: >100ms amber, >500ms red; packet loss >1% amber, >5% red; monthly Azure SLA review	6.6-D, 6.6-E
User Account Lifecycle	Onboarding (Day 1 setup, RBAC per role, MDM enrolled); role changes (<24hr); offboarding (same-day deactivation); quarterly access recertification	6.6-F
Asset & Licensing	Hardware refresh: laptops 4yr, servers 5yr, network 7yr; warranty tracking (60-day alert); quarterly licence compliance	6.6-G, 6.6-H
Succession Planning	ICT Officer → Asst ICT documented; quarterly skill rotation; knowledge transfer register; emergency WFP ETC/IFRC surge	6.6-I

10. Annex Toolkit Quick Reference

Group	Key Annexes	Count
SOP 6.1 Data Management	Data Receipt Register, Metadata Schema, Classification Matrix, Data Request Register, Backup Verification, Quality Form, Exchange Specs, Cross-SOP Data Flow, Stewardship Matrix, AAR Template	10
SOP 6.2 ICT Infrastructure	MNO Directory, Satellite Inventory, VPN Standards, Cloud Matrix, Security Baseline, AU/UN Credentials, MET Feed Specs, ETC Procedures, ETK Checklist, IR Procedures, Network Topology, Cross-Functional Matrix, PPDR Reference	13
SOP 6.3 Reporting	SITREP, Flash Update, Rapid Assessment, Dashboard Indicators, Quality Checklist, Cross-SOP Flow, AAR, Donor Report, Accessibility Checklist, Call Centre Setup	10
SOP 6.4 BCM & DR	Failover Checklist, OSOCC Readiness, Virtual OSOCC, DR Test Plan, Cloud Restore, SPOF Testing, Power Continuity Log, Post-Recovery Validation, BCM-ICT Joint Activation	9
SOP 6.5 Cybersecurity	IR Plan, Severity Matrix, Vuln Scanning Schedule, Pen Test Scoping, SIEM Triage, Patch Lifecycle, Endpoint Checklist, Training Programme, Audit Checklist, DLP Policy	10
SOP 6.6 Steady-State	Daily Ops Checklist, Help Desk Form/SLA, Change Request/CAB, Network Dashboard, Cloud SLA Template, User Account Lifecycle, Licence Register, Asset Lifecycle, Succession Plan	9

11. Key Performance Indicators

Indicator	Target	Frequency	SOP
Data Quality Score	≥95%	Weekly	6.1
Backup Success Rate	100%	Daily	6.1
RPO Compliance (EW data)	≤15 min	Continuous	6.1/6.2
Network Availability	≥99.5%	Continuous	6.2
SATCOM Readiness	100%	Monthly	6.2
SITREP Timeliness	≥95%	Per event	6.3
Dashboard Uptime	≥99.9%	Continuous	6.3
Accessibility Compliance	100%	Per publication	6.3
Failover Drill Success	100%	Quarterly	6.4
Power Continuity Test Pass	100%	Monthly	6.4
Critical Vuln Remediation <24hr	≥95%	Per event	6.5
Phishing Click Rate	<10%	Quarterly	6.5
Help Desk SLA Compliance	≥90%	Monthly	6.6
Daily Ops Checklist	100%	Daily	6.6

12. Risk Matrix and Cross-SOP Integration

Top risks trainees must know (BGYOR):

Risk	Rating	Key Mitigation
Cyber Attack	Red 15	SECaaS; EDR; SOC 24/7; TESS+ baseline; IR procedures
Power Outage (Nacala)	Red 16	UPS; dual generator; solar backup; cloud failover
Unpatched Vulnerability	Red 15	Weekly scanning; Critical <24hr; pen testing; SIEM
Data Breach	Amber 10	MFA; encryption; SOC monitoring; ISO 27001 controls
Local MNO Failure	Amber 12	Multi-carrier; Starlink/VSAT backup
SITREP Production Delay	Amber 12	Template automation; shift coverage; surge roster
Three-Tier Failover Failure	Amber 10	Quarterly drills; OSOCC tests; cloud restore drills
ICT Officer SPOF	Amber 12	Succession plan; quarterly skill rotation; cross-training

Volume 6 integrates with all SHOC SOP Volumes: Finance (IFMS data feeds), Early Warning (hazard data ingestion, CAP validation), Emergency Comms/Telecom (ETK deployment, PPDR frequencies, Tampere Convention), ERT (field data, OSOCC integration), HR (HRIS platform), Security (access control, SIEM), SCM (procurement systems, asset tracking), and BCM (failover architecture, continuity triggers).

13. Assessment and Competency Verification

Competency	Assessment Method	Minimum Standard
Data classification (5-tier)	Written test: classify 5 data items	Correct classification and handling for all 5
Data validation and CAP verification	Practical: validate 3 data packages including CAP message	All errors identified; CAP validated <30min
SATCOM deployment (ETK)	Practical: deploy Starlink/VSAT terminal	Connectivity achieved within 4hr target
SITREP production (10-section)	Produce SITREP from operational scenario data	All 10 sections complete; accessibility verified
Three-tier failover execution	Scenario: activate Tier 1–3 failover sequence	Correct tier at each stage; RTO targets met
Incident response (5-step)	Scenario: respond to P1 cybersecurity incident	Containment <1hr; all 5 steps completed
Vulnerability remediation	Identify and remediate 3 vulnerabilities from scan results	Critical <24hr; correct severity triage
Change management (Normal + Emergency)	Complete change request for both types	CAB process followed; rollback documented
Help desk SLA compliance	Manage 5 tickets across all priority levels	Correct priority assigned; SLA met for each
DR restore and backup verification	Execute cloud restore drill and backup test	Data integrity confirmed; RTO measured

Certification requirement: ICT Officers and IM staff must demonstrate competency across all ten areas before assuming operational roles. Quarterly DR drills, semi-annual cybersecurity exercises, and annual full BCM exercises provide the primary assessment vehicles. Results are recorded in the ICT competency register and reviewed by the Head of Operations.

— END OF TRAINING MANUAL —